

Artificial intelligence

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Course Overview

About Me

Name: Oybek Eraliev Maripjon O'g'li

Bachelor Degree: Fergana Polytechnic Institute,
Electrical Engineering, 2008~2012.

Masters Degree: Fergana Polytechnic Institute,
Electrical Engineering, 2012~2014.

Ph.D degree: Inha University (South Korea),
Future Vehicle Engineering, 2019~2024.

Currently: Associate Professor in IUT, 2024~now



Course Overview

Researches



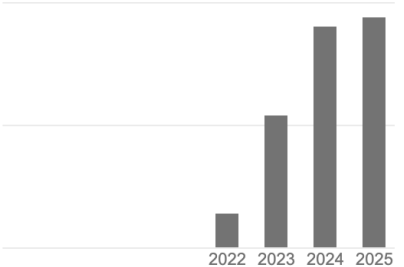
Oybek Eraliev 
PhD. Applied AI, ML Engineer. [Inha University](#)
Verified email at inha.edu
[Artificial Intelligent](#) [Machine/Deep Learning](#) [Diagnosis and Health monit...](#)



☐	TITLE	CITED BY	YEAR
☐	Sensing, perception, decision, planning and action of autonomous excavators OMU Eraliev, KH Lee, DY Shin, CH Lee Automation in Construction 141, 104428	76	2022
☐	Vibration-based loosening detection of a multi-bolt structure using machine learning algorithms O Eraliev, KH Lee, CH Lee Sensors 22 (3), 1210	38	2022
☐	Automatic optimization of one-dimensional CNN architecture for fault diagnosis of a hydraulic piston pump using genetic algorithm OEM Ugili, KH Lee, CH Lee IEEE Access 11, 68462-68472	22	2023
☐	Performance analysis of time series deep learning models for climate prediction in indoor hydroponic greenhouses at different time intervals O Eraliev, CH Lee Plants 12 (12), 2316	13	2023
☐	Experimental investigation on self-loosening of a bolted joint under cyclical temperature changes OMU Eraliev, YH Zhang, KH Lee, CH Lee Advances in Mechanical Engineering 13 (8), 16878140211039428	13	2021
☐	Self-loosening of a 3D-printed bolt by using three different materials under cyclical temperature changes O Eraliev, KH Lee, CH Lee Applied Sciences 12 (6), 3001	6	2022
☐	SCAE—Stacked convolutional autoencoder for fault diagnosis of a hydraulic piston pump with limited data samples O Eraliev, KH Lee, CH Lee Sensors 24 (14), 4661	5	2024
☐	Reliable fault diagnosis method based on an optimized deep belief network for gearbox O Eraliev, O Yakimov, CH Lee 드라이브·컨트롤 20 (4), 54-63	4	2023

Cited by

	All	Since 2020
Citations	177	177
h-index	6	6
i10-index	5	5



Year	Citations
2022	10
2023	35
2024	55
2025	65

Co-authors [EDIT](#)

No co-authors

Research Interests:
Applied AI, Machine/Deep Learning, NLP.



Course Overview

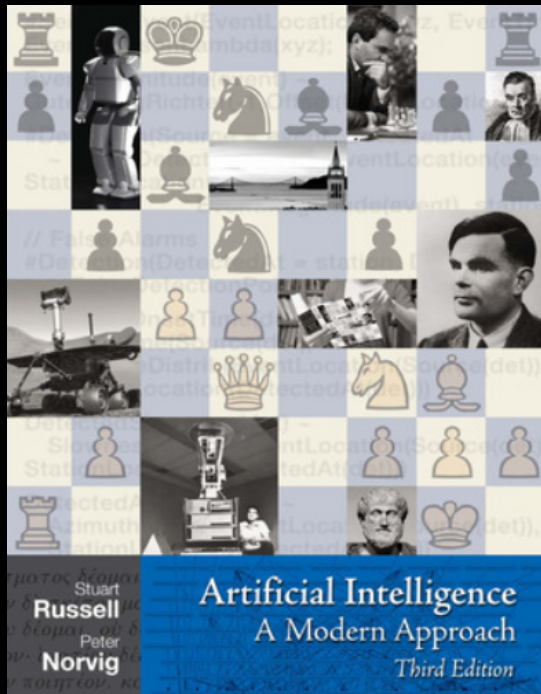
Researches



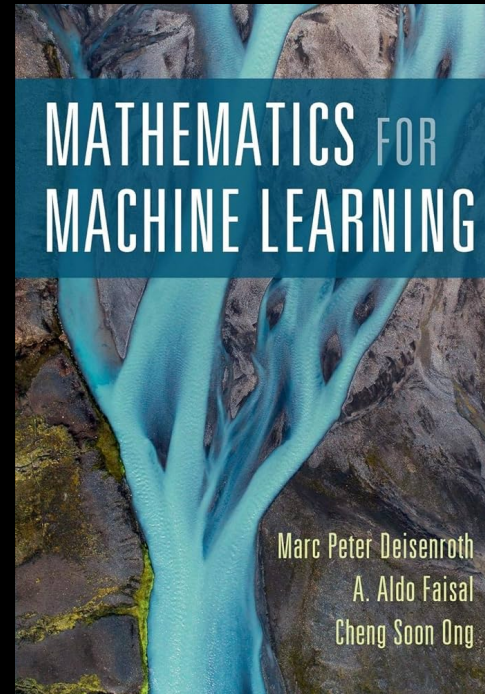
[@machine_learning_lab](https://t.me/@machine_learning_lab)

Course Overview

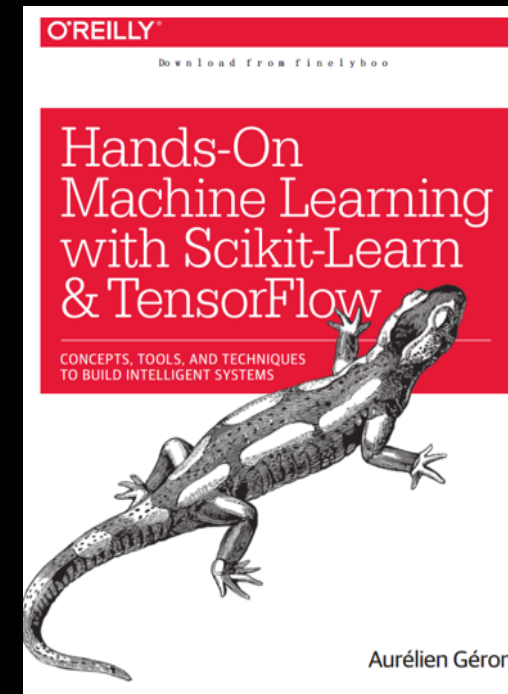
References



Artificial Intelligence



**Mathematics for
Machine Learning**



Machine learning

Course Overview

Class Progress and Evaluation

➤ Class progress

- PPT Lectures and discussion to achieve the objectives
- Attendance
- Exams
- Term Project (4~6 students in a team)

➤ Teaching Assistant and Office Hours

- Educational Assistant: Mr. Jasurbek Mamurov
- Office Hours: Tuesday & Thursday (14:00 ~ 15:30)

Course Overview

Class Progress and Evaluation

➤ Evaluation

- Attendance: 10%
 - medical certificates are not accepted
 - 8 and more absences automatically **Fail**
- Mid-Term Exams: 25%
- Final Exam: 25%
- Homework-Assignments: 20% (two 10% each)
- Term project: 20%
- **Extra points (optional): 10 points.**
 - Taking note (checking before mid exam and final exam)

Course Overview

Term Project

➤ Evaluation

- Make a team (4~6 students in a team)
- End-to-end AI project
- Choose a project topic (free topic, should be not the same with other teams), (Week 5)
- Topics should be in these fields: Computer Vision, NLP, Generative AI, etc.
- Make a project proposal (Week 6)
- Prepare a report of the project
- Representation of the project (Week 14, might be in Week 16).

Course Overview

Syllabus

- Week 1: Introduction of Artificial Intelligence
- Week 2: Core ML Algorithms
- Week 3: ML for Classification
- Week 4: ML for Clustering
- Week 5: Introduction to Deep Learning
- Week 6: Convolutional Neural Networks
- Week 7: Advanced CV Models
- Week 8: Mid-Term Exam

Course Overview

Syllabus

- Week 9: Natural Language Processing
- Week 10: NLP with Transformers
- Week 11: Generative Models
- Week 12: Reinforcement Learning
- Week 13: Deep Reinforcement Learning
- Week 14: Term Project Presentation
- Week 15: Final Exam



Introduction to Artificial Intelligence

Part – 1

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Welcome & Learning Objectives

By the end of this session, you will be able to:

- 1 **Define & Distinguish** — Explain what AI is, and how it relates to machine learning and deep learning.
- 2 **Explain Learning** — Describe how machines learn from data across three major ML paradigms.
- 3 **Describe Modern Systems** — Explain, at a conceptual level, how LLMs and computer vision models work.
- 4 **Evaluate Applications** — Assess real-world AI applications and their ethical implications.
- 5 **Discuss the Future** — Identify where the field is heading and what skills matter for your career.

AGENDA

PART 1

Foundations of AI

What AI is, its history, and how it relates to ML & DL.

PART 2

How Machines Learn

Supervised, unsupervised & reinforcement learning.

PART 3

Modern AI Systems & Applications

Vision, NLP, generative AI, and real-world uses.

PART 4

Responsible AI & The Future

Limitations, bias, safety, and where the field is heading.

Foundations of AI

What Is AI?

Artificial

made or produced by human beings rather than occurring naturally, especially as a copy of something natural.

Foundations of AI

What Is AI?

Intelligence

the ability to acquire and apply knowledge and skills.

Foundations of AI

What Is AI?

Artificial Intelligence is the field of building systems that perform tasks which normally require human intelligence — perceiving, reasoning, learning, and acting.

1

Perceive

Interpret images, sound & text

e.g., recognizing a face in a photo

2

Reason

Draw conclusions & solve problems

e.g., diagnosing a fault from symptoms

3

Learn

Improve performance from experience

e.g., a spam filter that gets sharper over time

4

Act

Take actions in the world

e.g., a self-driving car steering & braking

A Brief History of AI



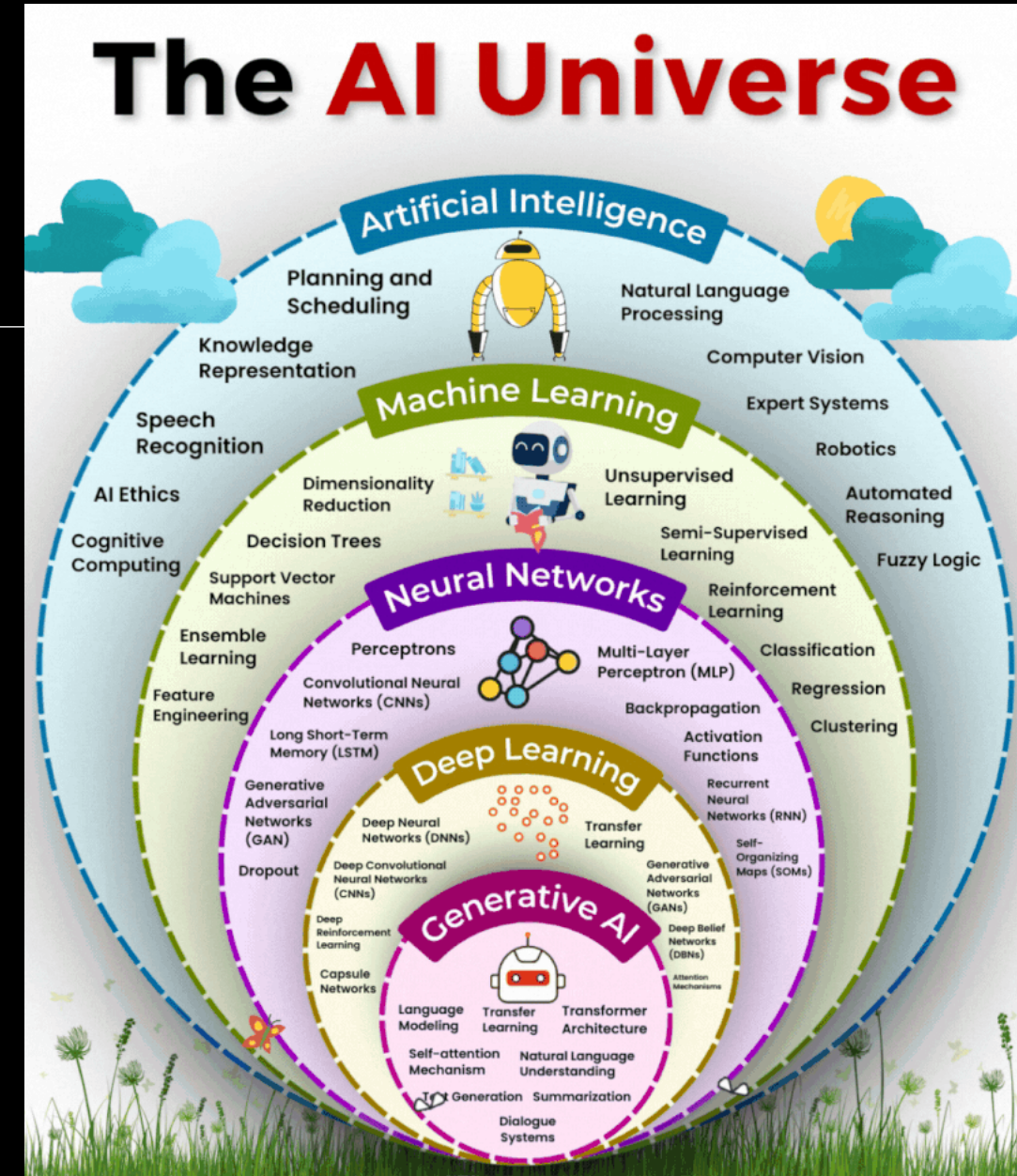
Foundations of AI

AI, Machine Learning & Deep Learning

AI - The broadest goal: building systems that behave intelligently — however that's achieved, including hand-coded rules.

Machine Learning - A subset of AI: systems that learn patterns from data instead of following explicit, hand-written rules.

Deep Learning - A subset of ML: uses multi-layer neural networks to learn increasingly abstract representations from data.



Foundations of AI

What Is AI?

➤ These six disciplines compose **most of AI**.

- 1 Natural language processing
- 2 Knowledge representation
- 3 Automated reasoning
- 4 Machine learning
- 5 Computer vision and speech recognition
- 6 Robotics

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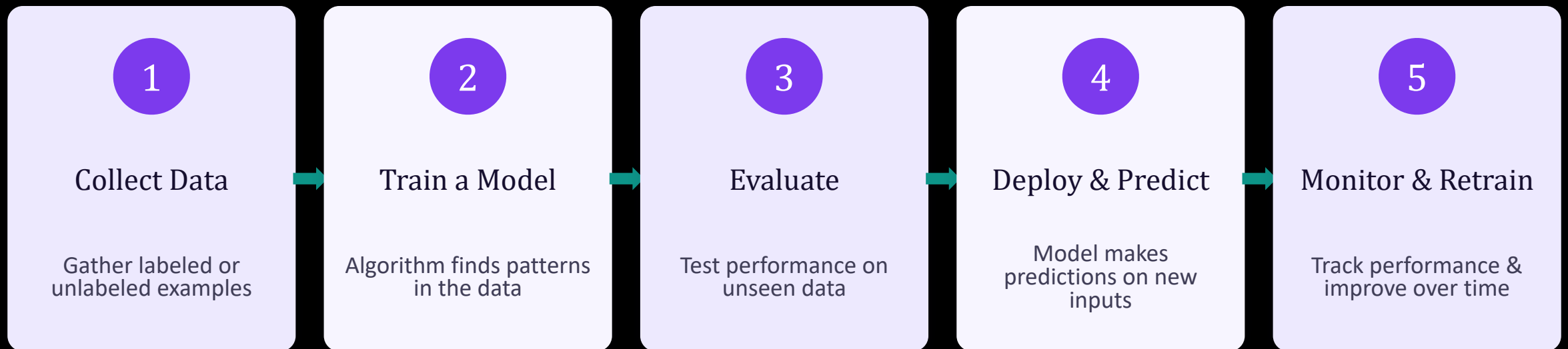
Vision, NLP, transformers, generative AI, and real-world uses.

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Responsible AI & The Future

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The Machine Learning Pipeline



Note: this loop repeats — production models are continuously monitored and retrained as new data arrives.

Supervised Learning

The model learns from labeled examples — inputs paired with the correct answer for each one.

HOW IT WORKS

Show many (input, correct output) pairs



Model learns the mapping between them



Model predicts labels for new, unseen inputs

EXAMPLES

Spam Detection

Email → spam or not spam, learned from millions of labeled messages.

Price Prediction

House features → sale price, learned from past sales data.

Medical Diagnosis

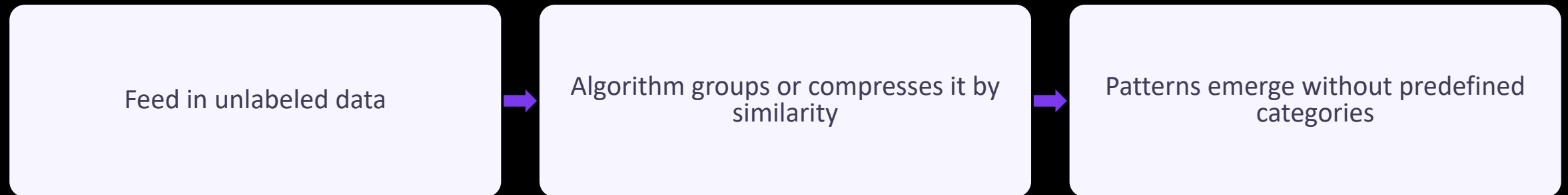
X-ray image → disease present or absent, learned from labeled scans.



Unsupervised Learning

The model finds patterns and structure in data that has no labels — there's no “correct answer” provided.

HOW IT WORKS



EXAMPLES

Customer Segmentation

Grouping shoppers by behavior for targeted marketing.

Anomaly Detection

Flagging unusual patterns in network traffic or transactions.

Topic Discovery

Finding themes across a large collection of documents.

Reinforcement Learning

An agent learns by taking actions in an environment and receiving rewards or penalties for the outcomes.

HOW IT WORKS

Agent observes the current state



Agent takes an action & receives a reward



Agent updates its strategy to maximize reward

EXAMPLES

Game-Playing Agents

AlphaGo and Atari-playing agents learning through trial and error.

Robotics

Robots learning to walk, balance, or grasp objects.

Resource Management

Optimizing data-center cooling to cut energy use.

Deep Learning: Why Now?

01

Data

Explosion of digital images, text & sensor data to train on.

02

Compute

GPUs & TPUs make training large networks practical.

03

Algorithms

Better architectures (CNNs, Transformers) & training tricks.

The idea is old: deep learning has existed in theory since the 1980s — it took all three of these coming together around 2012 to make it practical.



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Computer Vision

Computer vision teaches machines to interpret and understand visual information — images and video — from the world.

EXAMPLES IN PRACTICE

1

Facial Recognition

Unlocking phones and verifying identity from a face.

2

Medical Imaging

Detecting tumors or fractures in scans and X-rays.

3

Self-Driving Perception

Detecting pedestrians, lanes, and traffic signs in real time.

4

Quality Inspection

Spotting defects on manufacturing lines automatically.

Natural Language Processing (NLP)

NLP enables machines to understand, interpret, and generate human language — written or spoken.

EXAMPLES IN PRACTICE

1

Machine Translation

Google Translate converting text between languages.

2

Sentiment Analysis

Judging whether customer reviews are positive or negative.

3

Voice Assistants

Siri and Alexa understanding spoken requests.

4

Chatbots & Support

Automating customer service conversations.



Generative AI & Large Language Models

Generative AI creates new content — text, images, audio, code — instead of just classifying or predicting a label. Large Language Models (LLMs) like GPT are trained on massive text corpora to generate human-like language.

EXAMPLES IN PRACTICE

1

Text Generation

Chat assistants that answer questions and write drafts.

2

Image Generation

Diffusion models creating original images from a prompt.

3

Code Generation

Writing and completing code from natural language.

4

Summarization

Condensing long documents into key points.

How LLMs Actually Work (Simplified)

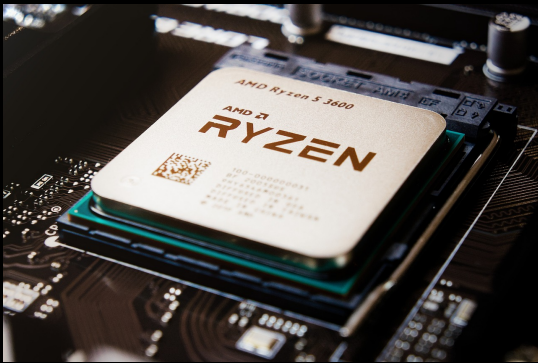


Important caveat: LLMs predict plausible text — they don't “know” facts the way a database does. That's why they can confidently state incorrect information, a failure mode known as “hallucination.”

The Foundations of Artificial Intelligence

Computer engineering

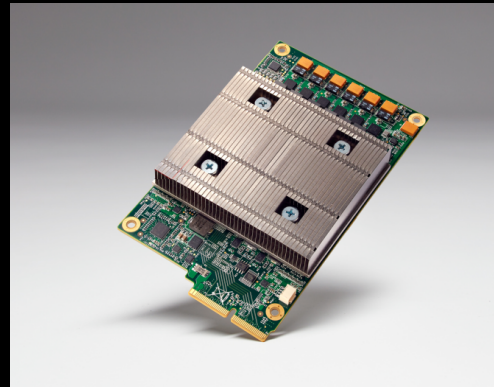
How can we build an efficient computer?



CPU



GPU

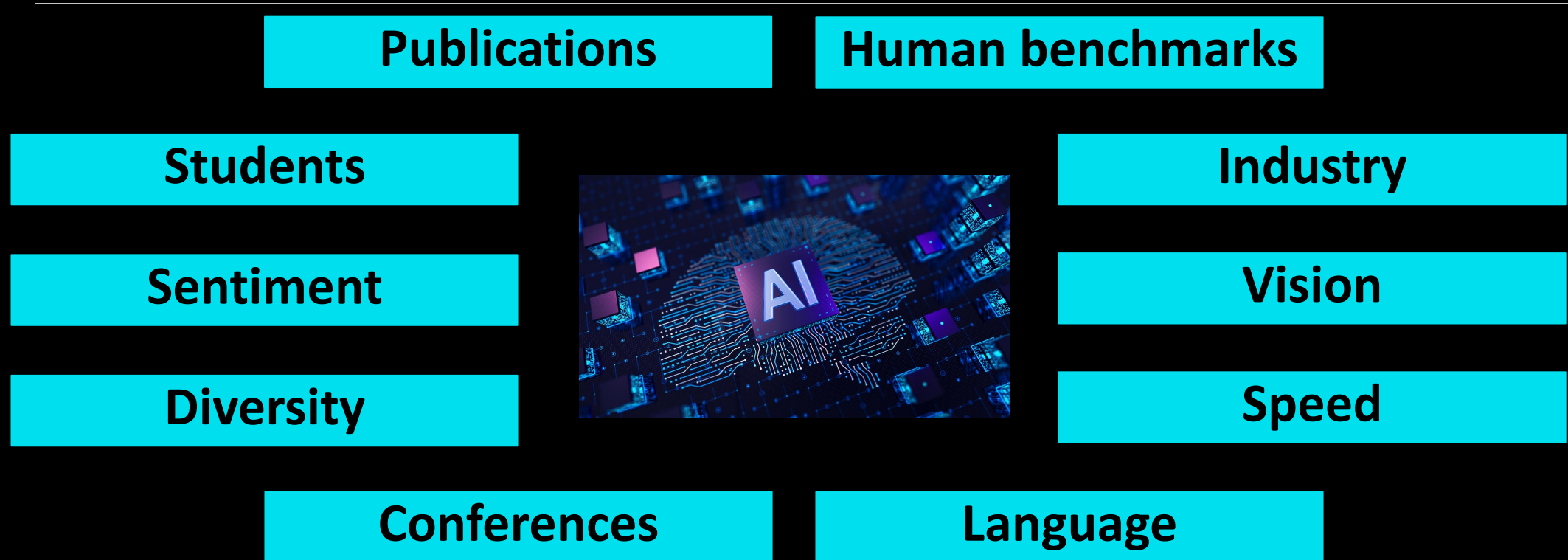


TPU



Quantum Computing

The State of the Art



The State of the Art

What can AI do today?

Robotic vehicles:

In 2018, Waymo test vehicles passed the landmark of 10 million miles driven on public roads without a serious accident, with the human driver stepping in to take over control only once every 6,000 miles.

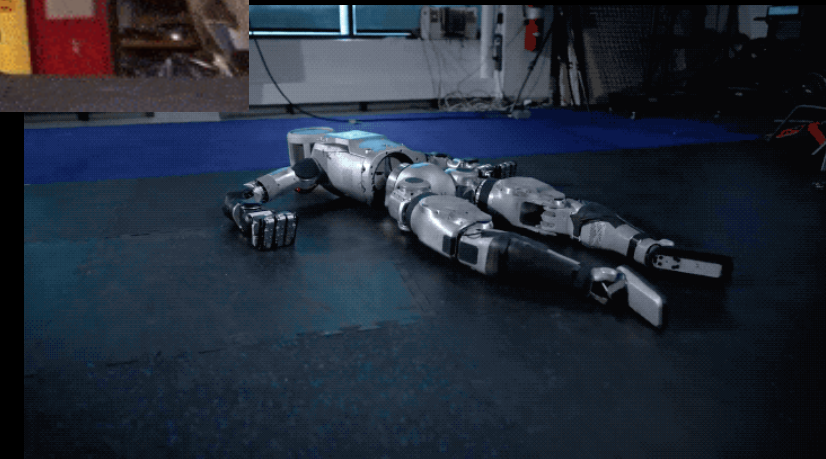


The State of the Art

What can AI do today?

Legged locomotion:

Atlas, a humanoid robot, not only walks on uneven terrain but jumps onto boxes and does backflips.

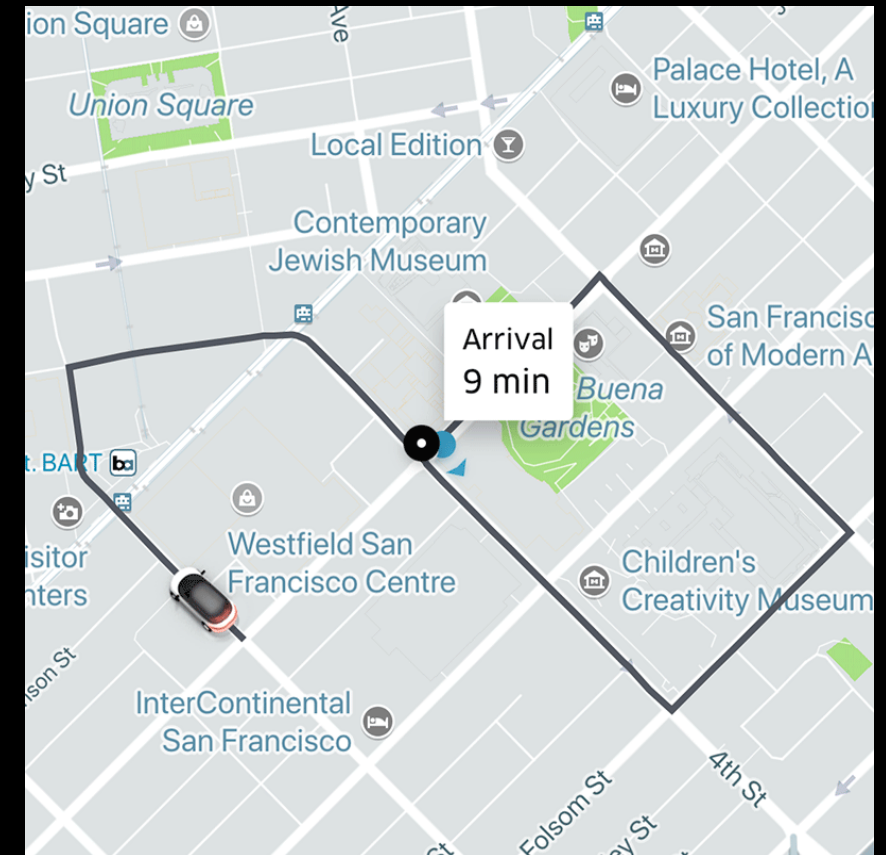


The State of the Art

What can AI do today?

Autonomous planning and scheduling:

Every day, ride hailing companies such as Uber, Yandex and mapping services such as Google Maps provide driving directions for hundreds of millions of users, quickly plotting an optimal route taking into account current and predicted future traffic conditions.

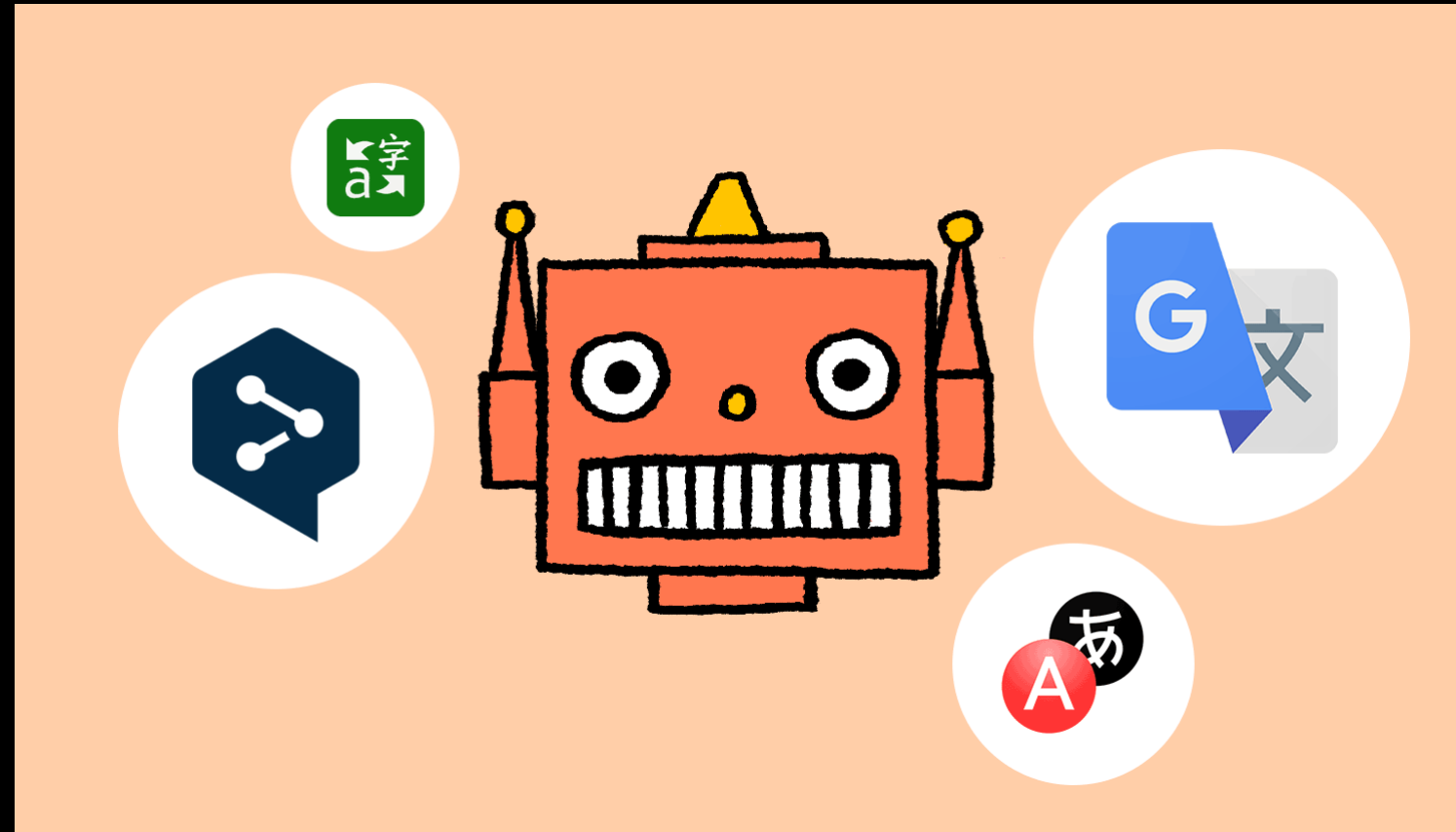


The State of the Art

What can AI do today?

Machine translation:

Online machine translation systems now enable the reading of documents in over 100 languages, including the native languages of over 99% of humans, and render hundreds of billions of words per day for hundreds of millions of users.



The State of the Art

What can AI do today?

Speech recognition:

Alexa, Siri, Cortana, and Google offer assistants that can answer questions and carry out tasks for the user.

For example, the Google Duplex service uses speech recognition and speech synthesis to make restaurant reservations for users, carrying out a fluent conversation on their behalf.

ALEXA vs SIRI vs GOOGLE



The State of the Art

What can AI do today?

Recommendations:

Companies such as Amazon, Facebook, Netflix, Spotify, YouTube, Walmart, and others use machine learning to recommend what you might like based on your past experiences and those of others like you.



The State of the Art

What can AI do today?

Medicine:

AI algorithms now equal or exceed expert doctors at diagnosing many conditions, particularly when the diagnosis is based on images. Examples include Alzheimer's disease (Ding et al., 2018), metastatic cancer (Liu et al., 2017; Esteva et al., 2017), ophthalmic disease (Gulshan et al., 2016), and skin diseases (Liu et al., 2019c).



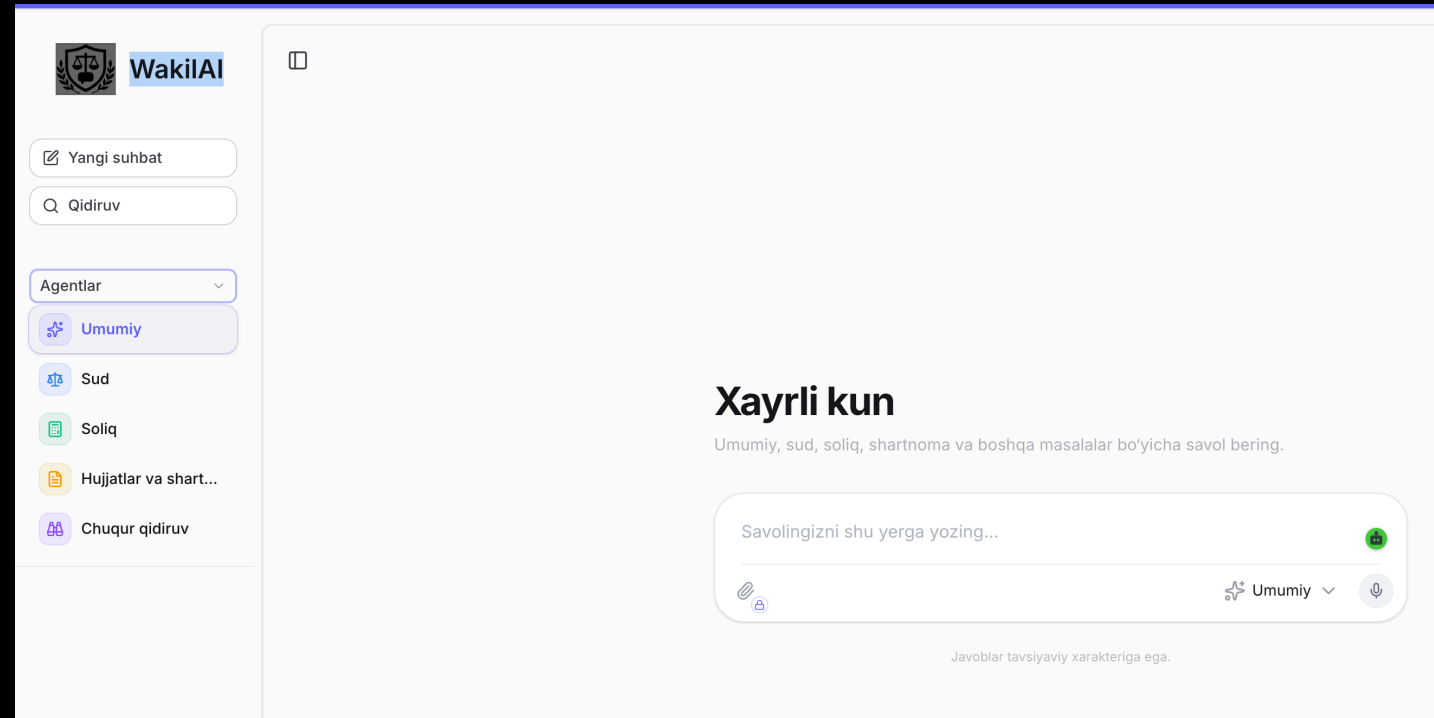
The State of the Art

What can AI do today?

Law:

Our national product

wakil.ai developed by HumblebeeAI



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Limitations, bias, safety, and where the field is heading.

Limitations of Today's AI

- 1 **Hallucination:** Models can generate confident, fluent statements that are simply false.
- 2 **Lack of True Understanding:** Systems perform pattern matching, not genuine comprehension or reasoning.
- 3 **Brittleness:** Small, unexpected changes to input can cause surprisingly large errors.
- 4 **Data Dependency:** Models inherit the biases, gaps, and blind spots of their training data.

Bias & Fairness in AI

AI systems can learn and amplify unfair patterns already present in their training data.

REAL-WORLD EXAMPLE

Facial Recognition

Historically shown higher error rates for women and people with darker skin tones, largely due to underrepresentation in training datasets.

REAL-WORLD EXAMPLE

Hiring Algorithms

Tools trained on historical resumes have downgraded candidates based on gender-correlated signals in the data.

Takeaway: fair AI requires diverse data, testing across subgroups, and ongoing auditing after deployment.

The Road Ahead: Trends to Watch

Multimodal AI

Models that combine text, image, audio, and video in one system.

Autonomous Agents

AI that plans and executes multi-step tasks with less supervision.

Smaller, Efficient Models

Capable models running locally on personal devices.

Growing Regulation

Frameworks like the EU AI Act shaping how AI is built & deployed.

Careers & Skills in AI

- 1 **Machine Learning Engineer** — Builds & deploys models into production systems.
- 2 **Data Scientist** — Analyzes data and builds predictive models.
- 3 **AI / ML Researcher** — Advances the state of the art in the field.
- 4 **AI Ethicist / Policy Analyst** — Evaluates the societal impact of AI systems.
- 5 **AI Product / Prompt Roles** — Applies AI capabilities to real products & workflows.

Skills worth building: programming (Python), statistics & linear algebra, and — increasingly — critical thinking about when and how to apply AI responsibly.

Key Takeaways

- ✓ AI is the broad goal; ML and DL are the primary tools we use to reach it today.
- ✓ Machines learn through supervised, unsupervised, and reinforcement paradigms.
- ✓ Transformers plus large-scale data & compute unlocked today's generative AI boom.
- ✓ AI creates enormous value — and real risks around bias, privacy, and misuse.
- ✓ Responsible use means understanding both what AI can do and where it falls short.



